

NASA Lunar In-Situ Resource Utilization Technology Overview

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Koorosh R. Araghi, NASA JSC ISRU Technology Manager
koorosh.r.araghi@nasa.gov

In Situ Resource Utilization (ISRU) Capability – ‘Prospect to Product’

ISRU involves any hardware or operation that harnesses and utilizes ‘in-situ’ resources to create commodities* for robotic and human exploration and space commercialization

Destination Reconnaissance & Resource Assessment

Assessment and mapping of physical, mineral, chemical, and water/volatile resources, terrain, geology, and environment

Resource Acquisition, Isolation, & Preparation

Atmosphere constituent collection, and soil/material collection via drilling, excavation, transfer, and/or manipulation before Processing

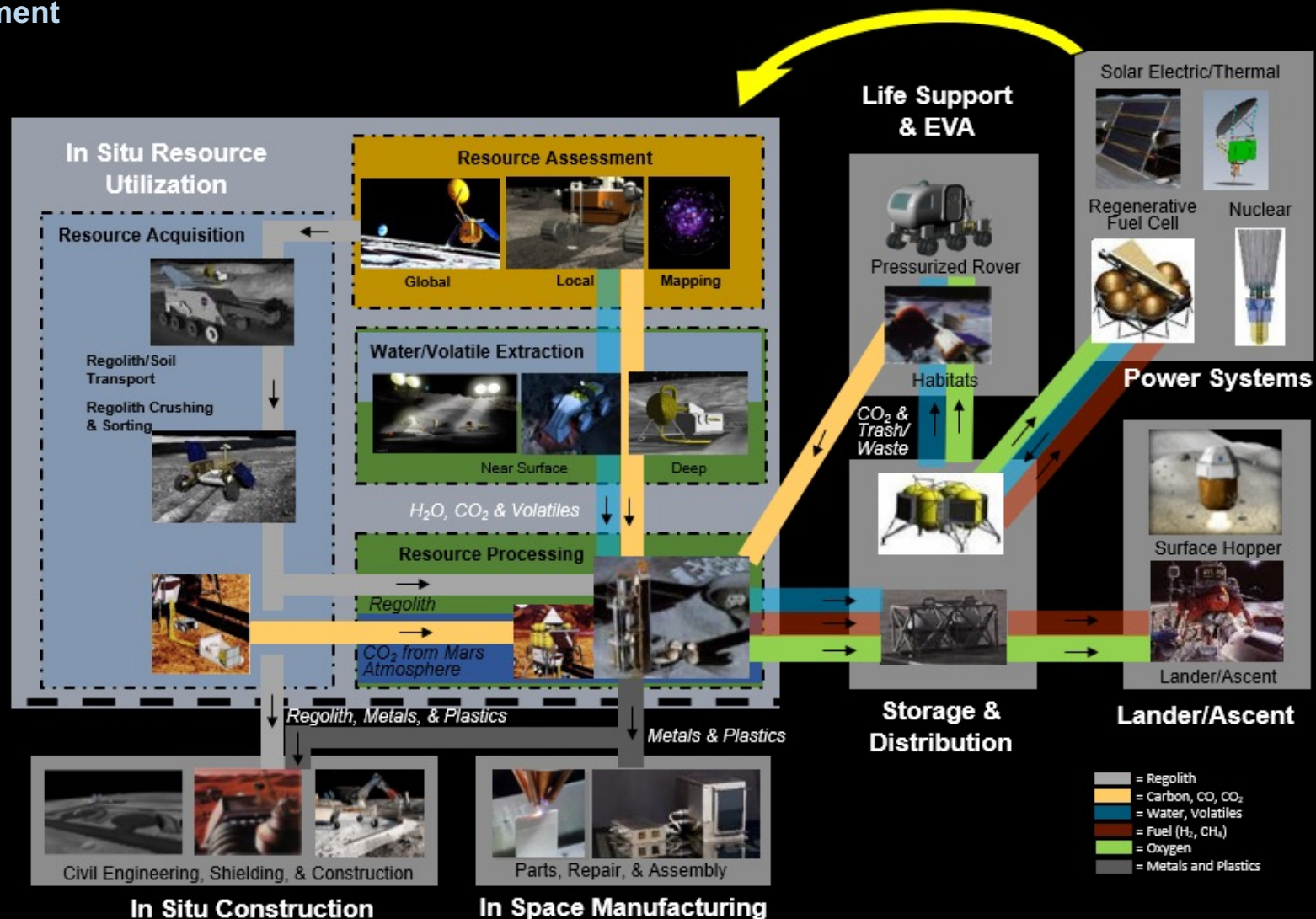
Resource Processing

Chemical, thermal, electrical, and or biological conversion of acquired resources and intermediate products into

- Mission Consumables
- Feedstock for Construction & Manufacturing

Water/Volatile Extraction

A subset of both Resource Acquisition and Processing focused on water and other volatiles that exist in extraterrestrial soils



➤ **ISRU is a capability involving multiple disciplines and elements to achieve final products**

➤ **ISRU does not exist on its own. It must link to users/customers of ISRU products**

Why Use Space Resources?

- **They can reduce mission and architecture mass and costs**

- **Launch mass savings:** 1 kg on Moon/Mars = 7.5 to 11 kg launched into Earth orbit
- **Reduce launch numbers:** lower mass = less launches (also less launches reduces mission risk)
- **Supports reuse of mission transportation assets:** reuse = lower cost; single stage landers possible
- **Supports terrestrial industry/Enables space commercialization:** reduces risk for industry involvement thereby reducing development and life cycle costs

- **They can increase safety for crew and mission success**

- **Ensure crew safety:** dissimilar redundancy for life support consumables, *in situ* repairs, radiation shielding, increased independence for Earth logistics
- **Provide critical solutions for mission assurance:** safe landing zones preparation, extra propellant for return to Earth (leakage/storage failure), protective shielding
- **Minimizes impact of shortfalls in other system performance :** launch and lander propulsion performance, life support system closure
- **Enhance crew psychological health:** provides knowledge and tools for self-reliance and survival; larger habitats

- **They can enhance or enable mission capabilities not possible without them**

- **Mission life extensions and enhancements:** local supply of critical consumables without waiting for new Earth supply, repairs with local materials, repurposing of end-of-life hardware, trash disposal (propellants and planetary protection),
- **Increased surface mobility and access:** surface hoppers, ISRU produced reactants for surface mobility fuel cell power, extend duration of exploration activities in extreme environments
- **Increased science:** more mass for science equipment, greater surface access, ISRU excavation

- **Learning to use Space Resources can help us on Earth**

- Renewable Energy/CO₂ Reduction, Recycling/Repurposing, Water cleanup, Environmentally-friendly mining and construction

NASA Lunar ISRU Purpose

Lunar ISRU To Sustain and Grow Human Lunar Surface Exploration

- Lunar Resource Characterization for Science and Prospecting
 - Provide ground-truth on physical, mineral, and volatile characteristics – provide geological context;
 - Test technologies to reduce risk for future extraction/mining
- **Mission Consumable Production (O₂, H₂O, Fuel):**
- Learn to Use Lunar Resources and ISRU for Sustained Operations
 - *In situ* manufacturing and construction feedstock and applications

Lunar ISRU To Reduce the Risk and Prepare for Human Mars Exploration

- Develop and demonstrate technologies and systems applicable to Mars
- Use Moon for operational experience and mission validation for Mars; Mission critical application
 - Regolith/soil excavation, transport, and processing to extract, collect, and clean water
 - Pre-deploy, remote activation and operation, autonomy, propellant transfer, landing with empty tanks
- Enable New Mission Capabilities with ISRU
 - Refuelable hoppers, enhanced shielding, common mission fluids and depots

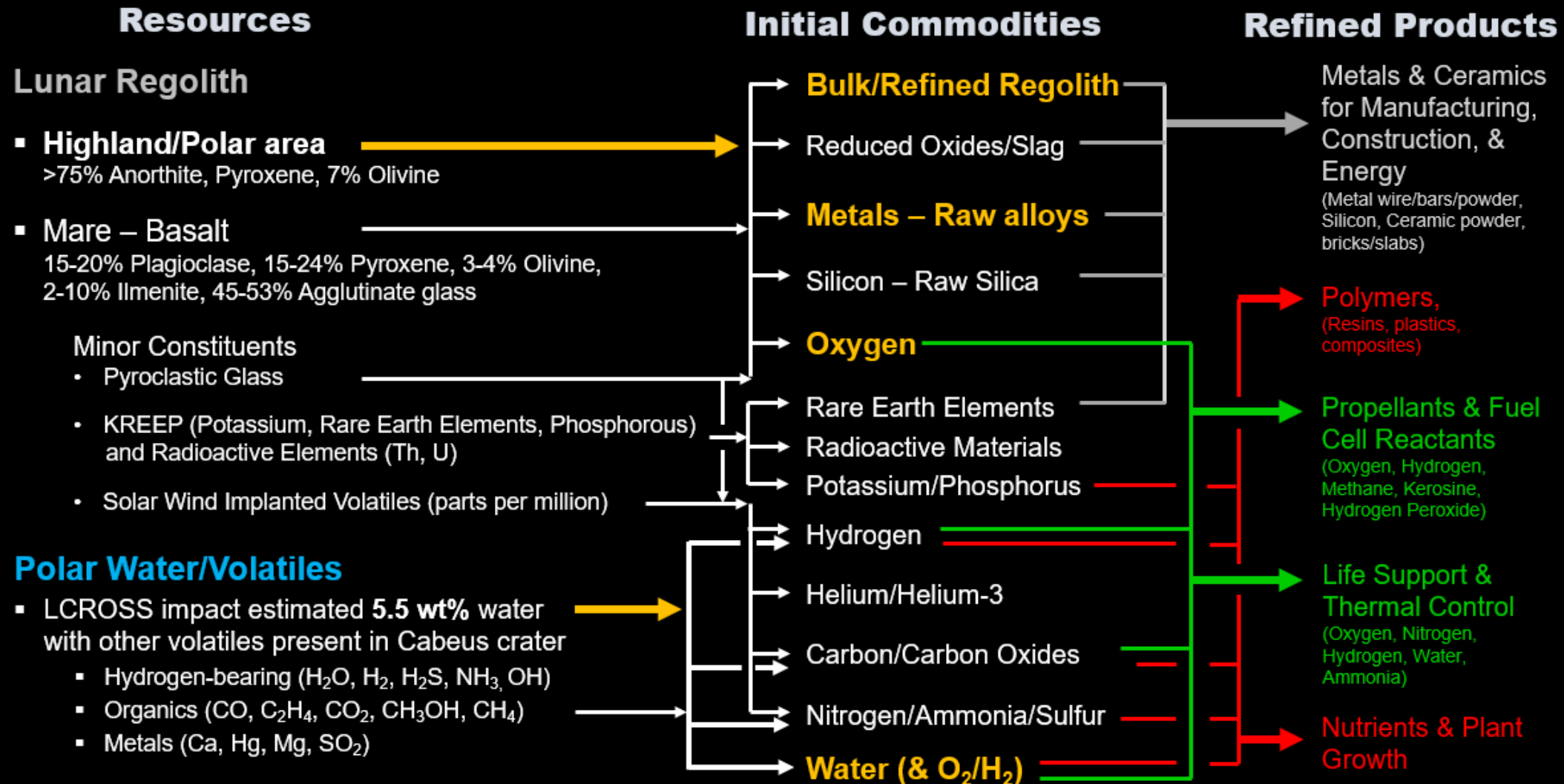
Lunar ISRU To Enable Economic Expansion into Space

- *SPD-1: Reinvigorating America's Human Space Exploration Program*
 - Promote International Partnerships
 - Promote Commercial Operations/Business Opportunities (Terrestrial and Space)
- Lunar Polar Water/Volatiles is Game Changing/Enabling
 - Support/promote establishment of reusable/commercial transportation

Lunar Resources and Commodities



- ISRU starts with the easiest resources to mine, requiring the minimum infrastructure, and providing immediate local usage
- The initial focus is on the lunar South Pole region (highland regolith and water/volatiles in shadowed regions)
 - ISRU will evolve to other locations, more specific minerals, more refined products, and delivery to other destinations



Gold/Bold text = most important initial commodities

Plan to Achieve ISRU Outcome

Scalable ISRU production/utilization capabilities including sustainable commodities on the lunar & Mars surface

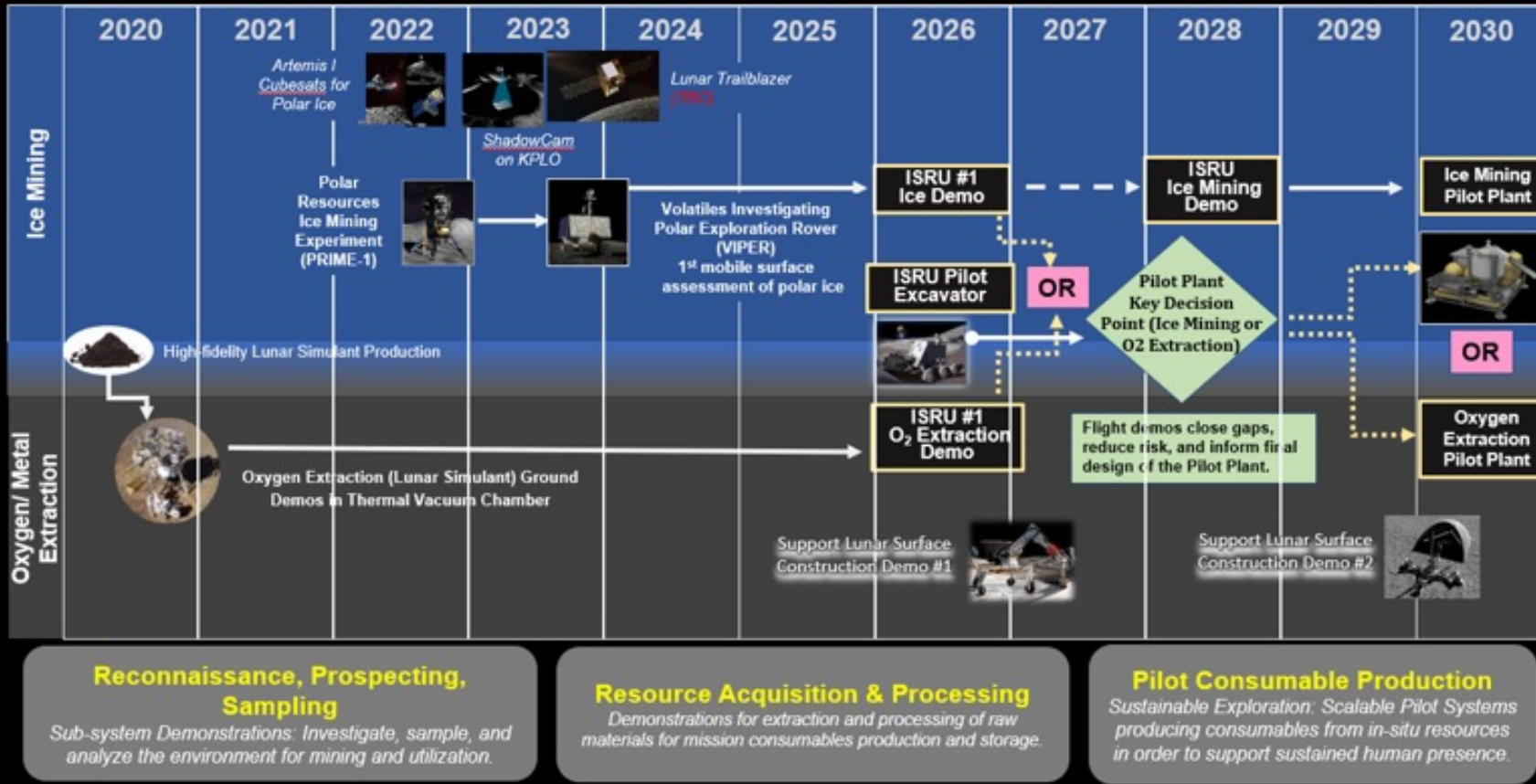


- **Know Customer Needs (Type and Quantity of Commodities) & Develop Suppliers**
 - Work with Artemis elements, Moon/Mars Surface Architecture, and International Partners
 - Work with Commodity users: Life Support & Food Production, Propulsion, Manufacturing, Construction
 - Understand all processing system wastes (life support, ISRU, manufacturing, construction) as potential new resource
 - Work with Terrestrial/Space Industry & Lunar Surface Innovation Consortium for Commercial Involvement & Opportunities
- **Perform Ground Development of Hardware and Systems until Ready for Lunar Flight**
 - Initiate a full range of ISRU & other discipline technologies across all TRLs (Technology Pipeline) to enable ISRU capabilities
 - Perform gravity related research (short duration & ISS) on material handling, resource processing, and feedstock behavior
 - Integrate lunar ISRU technologies and subsystems into systems for environmental and operational testing
 - Develop lunar ISRU components, subsystems, and operations (including autonomy) applicable to Mars ISRU systems
 - Engage Industry, Academia, and the Public to lay the foundation for long-term lunar economic development
- **Reduce Risk of ISRU for Human Exploration & Space Commercialization thru CLPS Missions**
 - Understand lunar polar resources for technology development, site selection, mission planning (SMD and ESDMD)
 - Obtain critical data (ex. regolith properties, validate feasibility of ISRU processes)
 - Demonstrate critical ISRU technologies in lunar environment, especially those that interact with and process regolith
- **Perform End-to-End ISRU Production of Commodities & Demonstrate Usage**
 - Production at sufficient scale to eliminate risk of Full-scale system
 - Initially use ISRU-derived commodity in non-mission critical application; examples include non-crewed ascent vehicle or hopper, extra fuel cell power, extra crew and EVA oxygen, construction demonstration, etc.
 - Involve industry in ISRU Demos and Pilot Plant to transition to Full-scale commercial operations
- **ISRU must be demonstrated on the Moon before mission-critical applications are possible**
 - NASA STMD is breaking the 'Chicken & Egg' cycle of past ISRU development priority and architecture insertion issues by developing and flying ISRU demonstrations and capabilities to the Pilot Plant phase

ISRU Path to Full Implementation & Commercialization*



**Proposed missions are contingent on appropriations and technology advancement*



**Full-scale
implementation
& Commercial
Operations
(see next chart)**

- Dual Path that includes both Water Mining and Oxygen/Metal from Regolith
 - O₂/Metal Path supports Surface Construction as well
- Ground development of multiple critical technologies in both pathways underway to maximize success and industry involvement
- Resource assessment missions to obtain critical data on mineral and water/volatile resources have started
 - PRIME-1 validates critical VIPER instruments and lunar highland material properties (for subsequent ground development)
- Demonstrations are aimed at reducing the risk of Pilot Plant design and operation (and subsequent Full-scale implementation)
 - Pilot Plant demonstrates performance, end-to-end operations, and quality of product for implementation and use

NASA ISRU Capability State of the Art and Current Work

Resource Assessment – Flight Development (TRL 4-6)

- Multiple instruments under development by SMD and STMD for resource collection and assessment
- Instruments to be flown on CLPS missions – PRIME-1 and VIPER for lunar ice characterization

Water Mining – Proof of Concept (TRL 2/3)

- 3 mining approaches and 6 water extraction technologies under development
- Challenges: Space Robotic, Break the Ice Lunar

Oxygen Extraction from Regolith – Engineering Breadboards/Field Test Units (TRL 4/5)

- Two Hydrogen Reduction systems built and tested at Pilot scale; terrestrial operations, non-flight mass/power, mare regolith, days/weeks operation (2008)
- Carbothermal Reduction system with solar concentrator built and tested as Sub-Pilot scale; terrestrial operations, mare regolith, non-flight mass/power, days/weeks operation (2010)
- Carbothermal & Hydrogen Plasma Oxygen extraction methods now reducing Highland simulants under laboratory conditions (TRL 3)

Oxygen/Metal Extraction from Regolith – Laboratory Proof of Concept

- Laboratory type/scale hardware: Molten Regolith Electrolysis (TRL 3/4); Ionic Liquid Reduction (TRL 2/3); International development of Molten Salt Electrolysis-ESA (TRL 3/4) and MRE-Israel (TRL 3/4)
- Bio-mining for oxygen/metal extraction (TRL 2/3)

Construction Feedstock (Low TRL: 2-4)

- Feedstock (blends of simulant and plastic) used in manufacturing & construction lab. demonstrations
- Mars concrete and soil/binders demonstrated: ACME & 3D Hab, Construction Centennial Challenge
- Size sorted lunar simulants being used for sintering construction tests
- Ilmenite beneficiation demonstrated on lunar-g aircraft
- 3D printer with simulant feedstock was tested on the ISS in the Additive manufacturing Facility
- Trash-to-Gas as start to conversion to fuels/plastics

Cross Cutting/System Level Resources

- 9 water electrolysis projects in 3 different types (PEM, SOE, Alkaline)
- NASA lunar simulant project initiated; Highland regolith simulant characterization & limited production
- External simulants available for purchase
- NASA Large dirty vacuum chamber almost ready at JSC; 2nd chamber at MSFC being modified

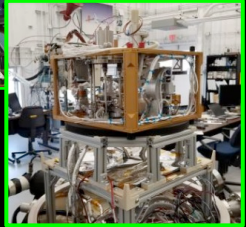
VIPER



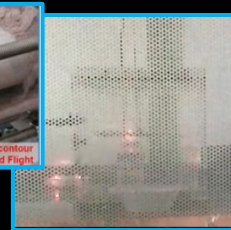
PRIME-1



LightWAVE



Lunar Auger Dryer
ISRU (LADI)



Radiant Gas
Dynamic Mining



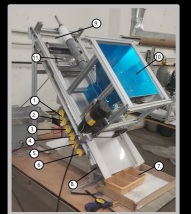
PILOT H₂
Reduction



Molten Regolith
Electrolysis



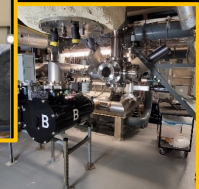
Carbothermal
Reduction



MMOST
Size
Separation &
Beneficiation



JSC 15' Dia. Dirty TVac



Simulants



Lunar Science & Resource Prospecting

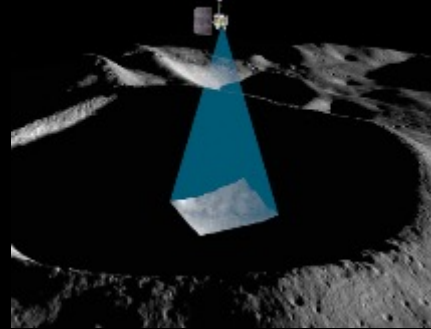


Orbital Missions

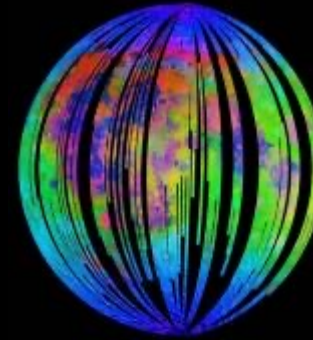
Lunar Reconnaissance Orbiter



ShadowCam on Korean Pathfinder Lunar Orbiter



Lunar Trailblazer
(Phase A)



Lunar IceCube



Lunar
Flashlight



LunaH-Map



Surface Missions

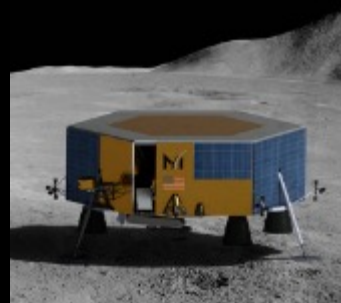
Astrobotics



Intuitive
Machines



Masten



PRIME-1

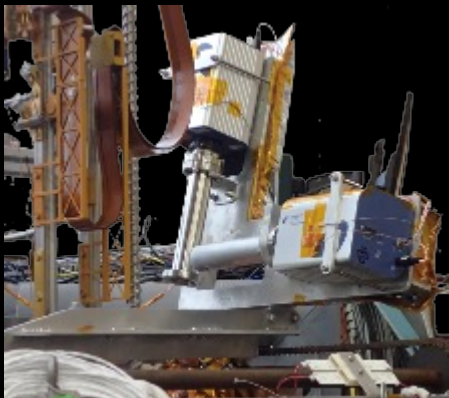


VIPER



PRIME-1 & VIPER

First Steps toward surface understanding of Polar Water and Volatiles



Polar Resources Ice Mining Experiment (Prime-1) on CLPS



Volatiles Investigation Polar Exploration Rover (VIPER)

- 2022 CLPS mounted payload to detect volatiles at 1-m depths
- Instruments include:
 - Near InfraRed Volatiles Spectrometer System (NIRVSS)
 - Mass Spectrometer Observing Lunar Operations (MSolo)
 - The Regolith and Ice Drill for Exploring New Terrain (TRIDENT)

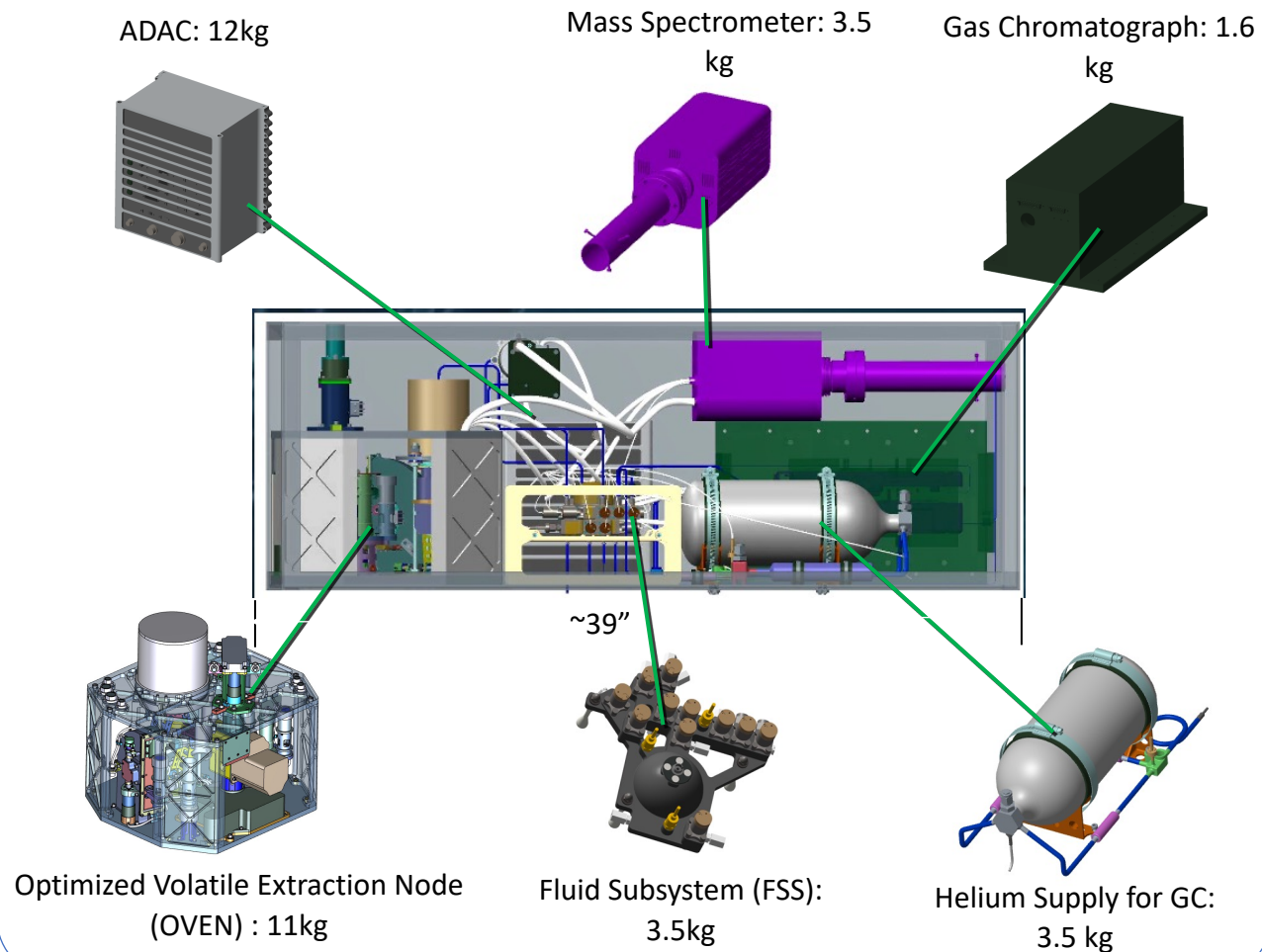


- 2023 at South Pole
- Measure volatiles at the lunar poles and acquire new key data on lateral and vertical distribution
 - Neutron Spectrometer System (NSS)
 - NIRVSS IR Spec
 - MSolo Mass Spec
 - TRIDENT Drill
- Build lunar resource maps for future exploration sites
 - Long duration operation (months)
 - Traverse 10's km

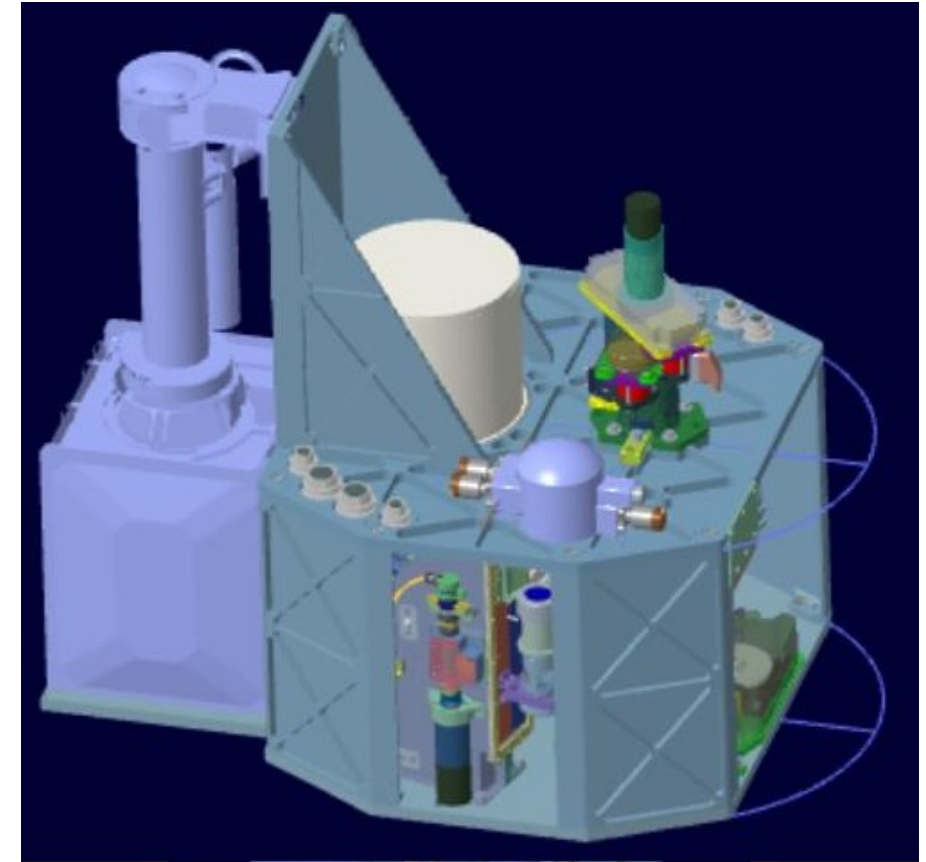
Light WAVE Prototype Concept



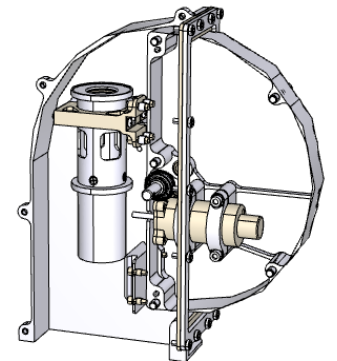
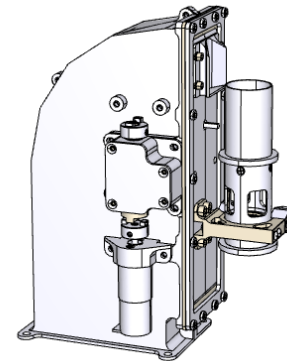
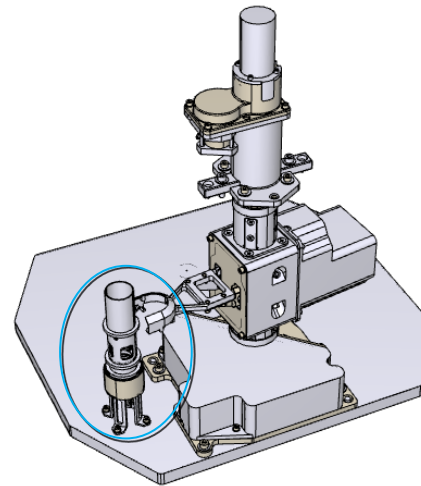
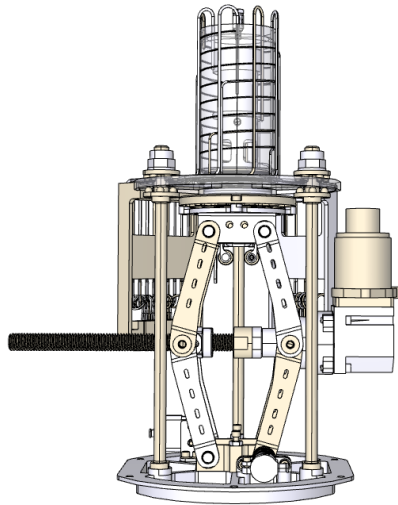
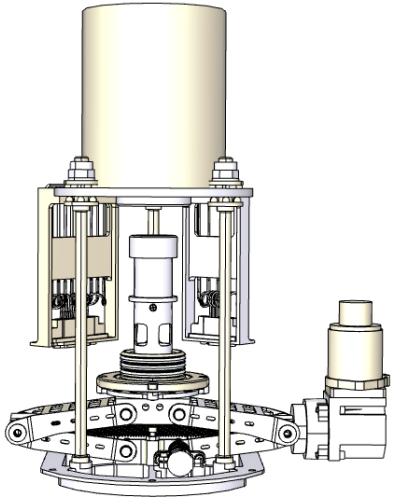
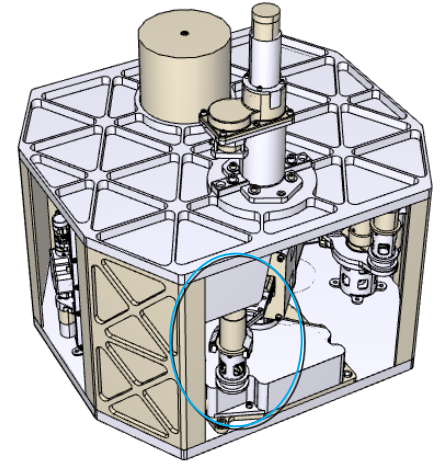
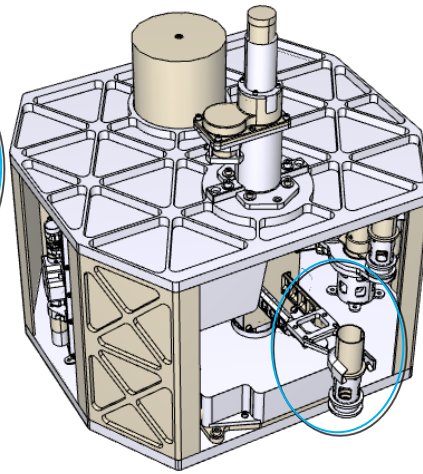
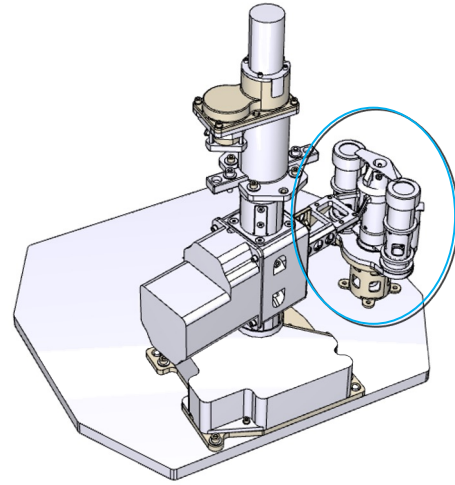
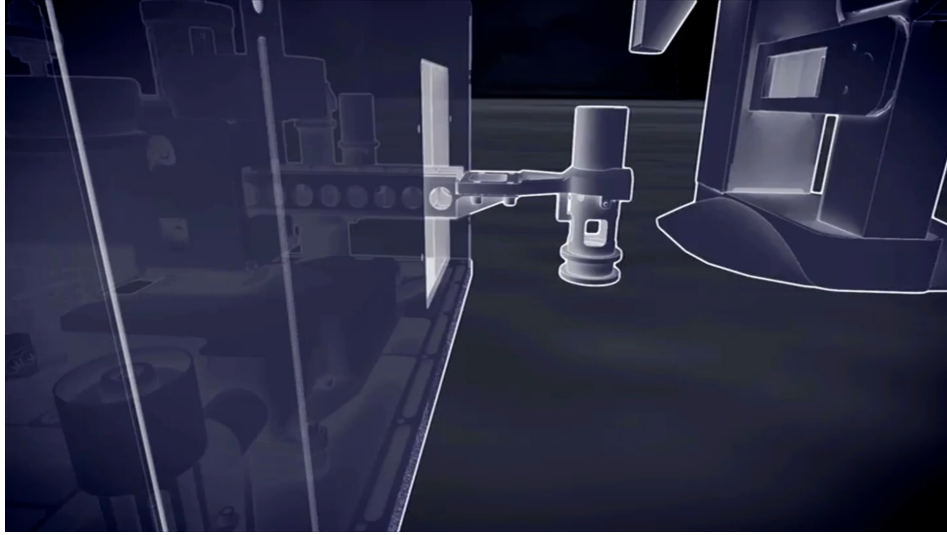
Original WAVE design at project closeout



Current Light WAVE CAD model



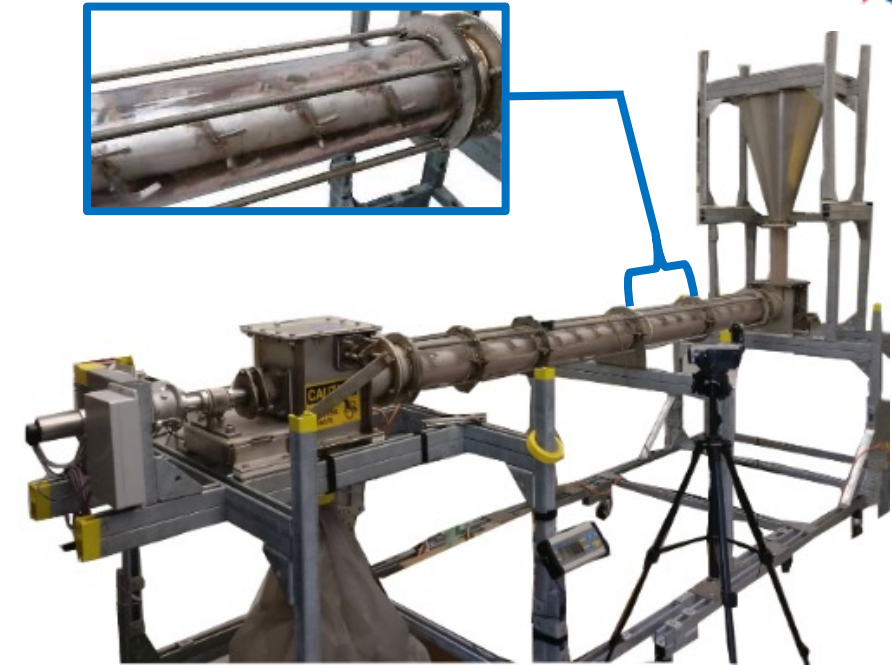
Light WAVE Concept of Operations



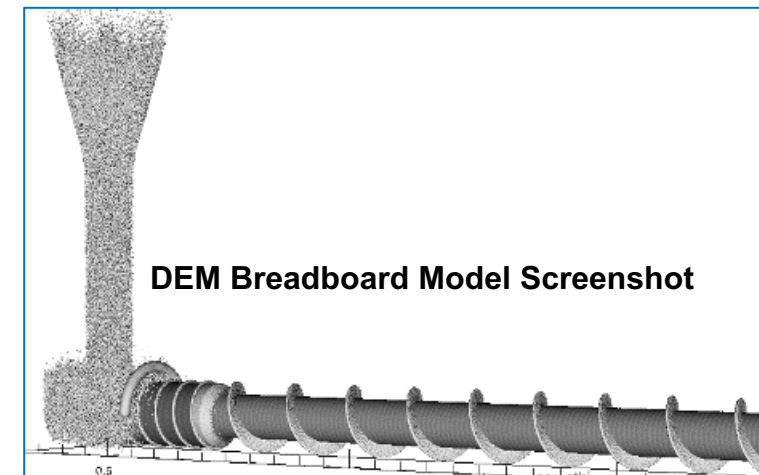
Lunar Auger Dryer ISRU (LADI)



- Technology Product Capability:
 - Continuously process water and volatiles from icy regolith in a lunar PSR
 - Process sublimates ice (thermally) and independent of ice type and distribution
- Technical Capabilities:
 - Breadboard test stand fully instrumented for full-scale flow
 - Two Breadboard casing materials available:
 - Plastic casing allows observation of plug formation and dust effects while measuring torque and RPM
 - Stainless Steel casing includes internal temp measurements and heaters controllable from 3 independent zones
- Exploration & Science Applicability:
 - Lunar: A screw conveyor dryer system operating from inside a PSR can produce water for both breathable air and propellant.
 - Mars: A screw conveyor dryer can be used on Mars



**Lunar Screw Conveyor Dryer
Breadboard Test Stand (clear casing installed)**



DEM Breadboard Model Screenshot

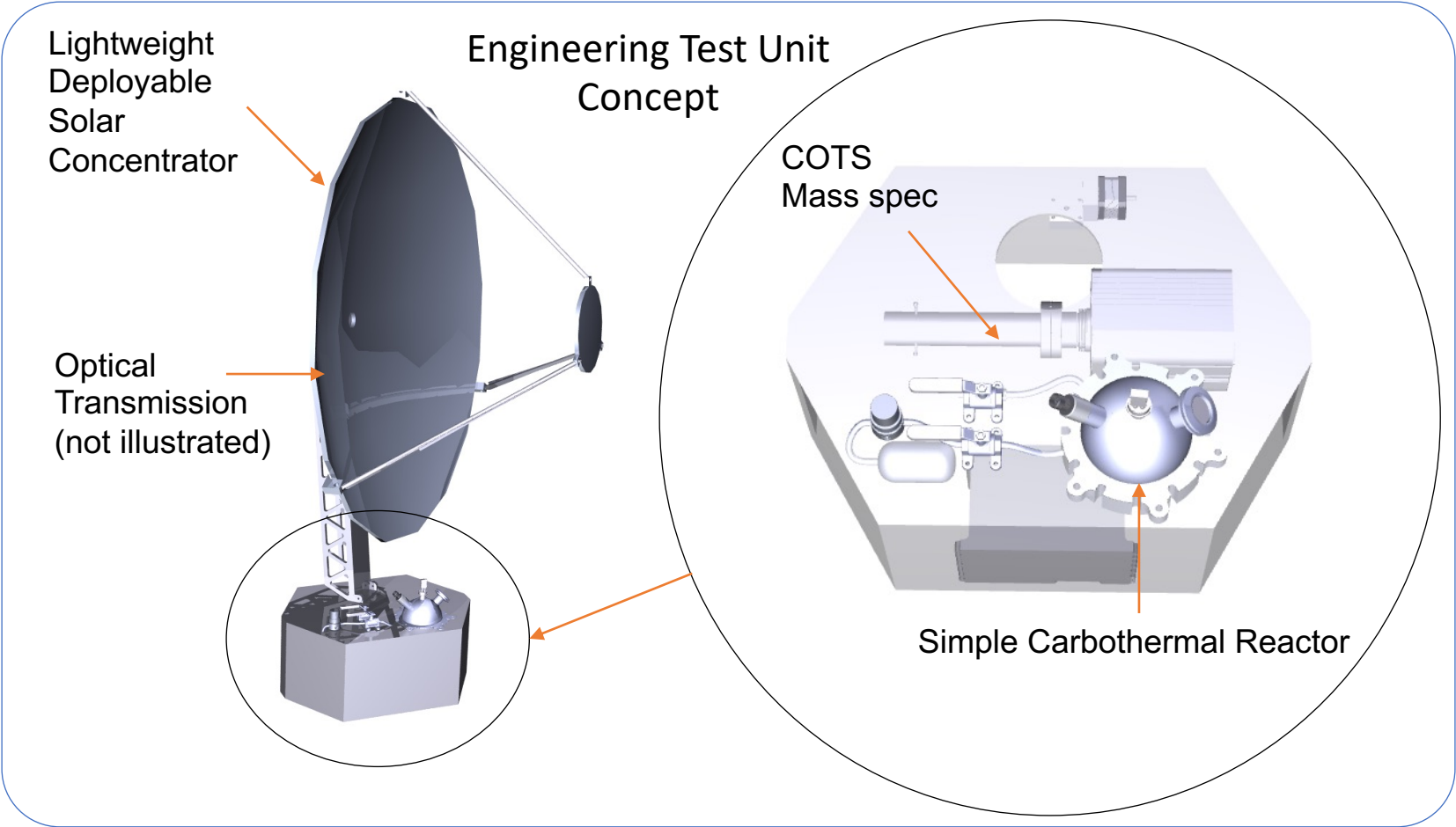


Carbothermal Reduction Demonstration (CaRD) Concept

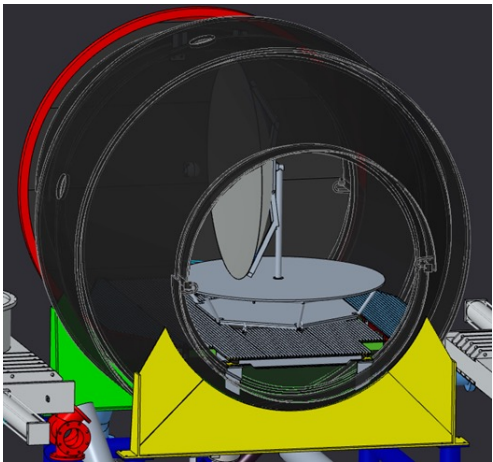


Technical Description: Carbothermal Reduction Demonstration (CaRD)

- CaRD is a ground development effort that can easily transition to becoming a CLPS payload in order to collect critical data for ISRU on the lunar surface.
- Concentrated solar energy melts lunar regolith and methane gas is used to remove oxygen from silicate minerals by Carbothermal reduction.
- Oxygen is extracted from the regolith in the form of carbon monoxide and carbon dioxide. All products of the reaction will be monitored using a mass spectrometer.
- Data collected on the lunar surface will inform a pilot plant design



Engineering Unit TRL5 in FY22	Prototype Unit TRL6 in FY23
Automatically deployed solar concentrator	Automatically deployed solar concentrator
Simple Carbothermal Reactor (Internally funded)	Automated Carbothermal Reactor
COTS Mass Spectrometer	Msolo (Flight Mass Spec)
COTS Electronics	Avionics
LabVIEW Software	Flight-Forward Software

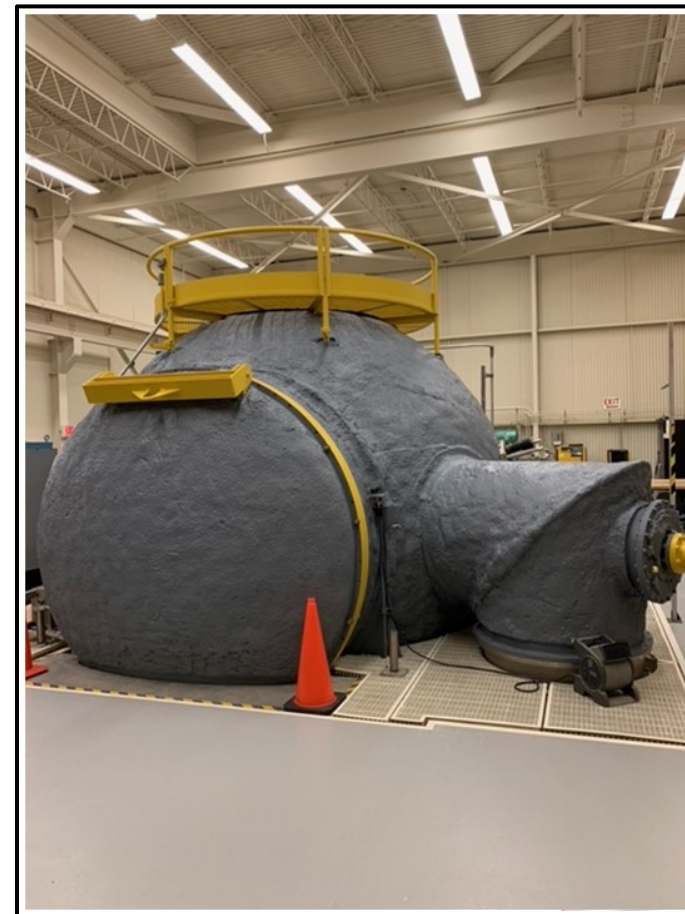


CaRD preliminary CAD model inside 15' chamber shroud (outer shell CAD removed)

Dusty Lunar Surface Simulation Capabilities Overview – Thermal Vacuum Chamber



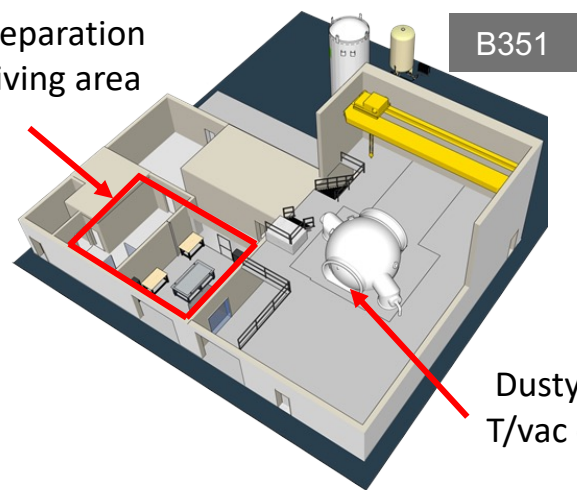
- Vacuum: 10^{-6} torr range (ultimate),
 - Pumping, rotary piston pumps, roots blowers & two 36" diameter diffusion pumps
- Atmosphere: Air or GN2 Re-pressurization
- Thermal Range: -185 °C to +120 °C (Thermal shroud temperatures)
- Internal Volume: $\sim 2,250 \text{ ft}^3$ (63.7 m^3)
- High channel count data feedthroughs
- High power feedthroughs available
- Max platform load: 4,000 lb (1814.36 kg)
- Test article operation on 48" x 96" regolith bed in chamber with surface interactions with regolith to depths down to ~ 8 in



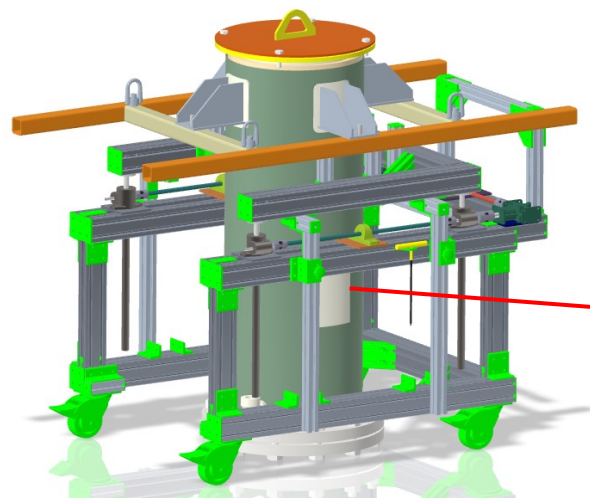
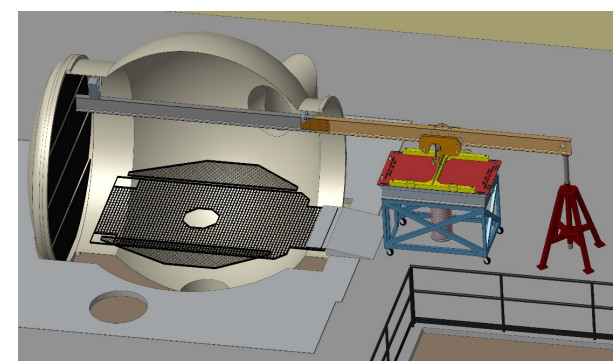
Dusty Lunar Surface Simulation Chamber



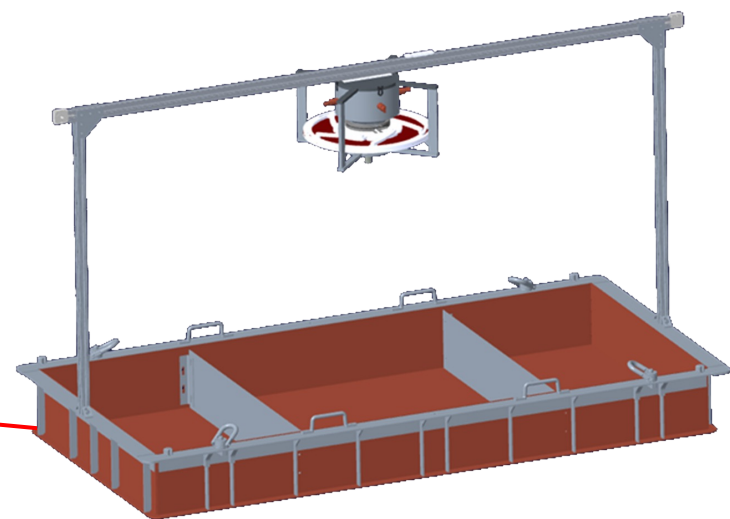
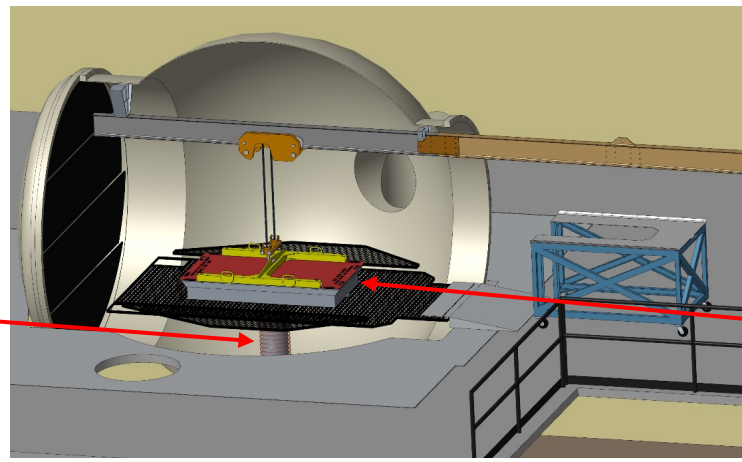
Lunar regolith preparation
& Hardware receiving area



Dusty Lunar Lunar
T/vac chamber



Lunar Icy drill Tube



Dust deposition system assembly
mounted on simulant bin

ISRU Development and Implementation Challenges



Space Resource Challenges 1

- R1 What resources exist at the site of exploration that can be used?**
- R2 What are the uncertainties associated with these resources?**
Form, amount, distribution, contaminants, terrain
- R3 How to address planetary protection requirements?**
Forward contamination/sterilization, operating in a special region, creating a special region

ISRU Operation Challenges 2

- O1 How to operate in extreme environments?**
Temperature, pressure/vacuum, dust, radiation, grounding
- O2 How to operate in low gravity or micro-gravity environments?**
Drill/excavation force vs mass, soil/liquid motion, thermal convection/radiation
- O3 How to achieve long duration, autonomous operation and failure recovery?**
No crew, non-continuous monitoring, time delay
- O4 How to survive and operate after long duration dormancy or repeated start/stop cycles with lunar sun/shadow cycles?**
'Stall' water, lubricants, thermal cycles

ISRU Technical Challenges

- T1 Is it technically and economically feasible to collect, extract, and process the resource?**
Energy, Life, Performance
- T2 How to achieve high reliability and minimal maintenance requirements?**
Thermal cycles, mechanisms/pumps, sensors/ calibration, wear

ISRU Integration Challenges 3

- I1 How are other systems designed to incorporate ISRU products?**
- I2 How to optimize at the architectural level rather than the system level?**
- I3 How to manage the physical interfaces and interactions between ISRU and other systems?**

Scale up, Long-duration, & Environmental testing with Realistic simulants Required

ISRU Capability Gaps to Achieve Initial Full-Scale Production*



*Estimates from Internal NASA and APL Lunar Surface Innovation Consortium Supply/Demand Workshop 9/17/2020)

Resource Assessment (Lunar Water/Ice) Capability Gaps

- Surface features and geotechnical data on regolith outside and inside permanently shadowed craters (PSRs)
- Understanding of water and contaminants as a function of depth and areal distribution
- Understanding of subsurface water/volatile release with heating
- Resolution of hydrogen and subsurface ice at <10s m scale (or less) for economic assessment & mine planning (orbital/surface)
- Instrument for polar regolith sample heating and released volatile characterization (minimum loss during transfer/evaluation)

Water Mining Capability Gaps

- Feasibility and operation of downhole ice/water vaporization and collection in cold-trap under lunar PSR conditions
- Feasibility and operation icy regolith transfer (low loss) and processing in reactor under lunar PSR conditions; min. 15,000 kg/yr; 3 years nom.
- Water and other volatile capture and separation; contaminant removal
- Electrical power & Thermal energy in PSRs for ice mining/processing (10s of KWs) – [Power System Gap](#)

Oxygen Extraction Capability Gaps

- Industrial-scale of regolith processing for oxygen (minimum of 10 mT O₂/yr; 3 years nom. with min./no maintenance)
- Regenerative oxygen & product gas clean-up (10,000 kg/yr)
- Measuring mineral properties/oxygen content before and after processing

Manufacturing & Construction Feedstock Capability Gaps

- Metal and metal alloy extraction from regolith: Post oxygen extraction or separate/multi-step refining
- Crushing, size sorting and mineral beneficiation of 100s mT per project for extraction and manufacturing/construction feedstock
- Production of 10s mT per project of plastic/binders and cement for manufacturing and construction

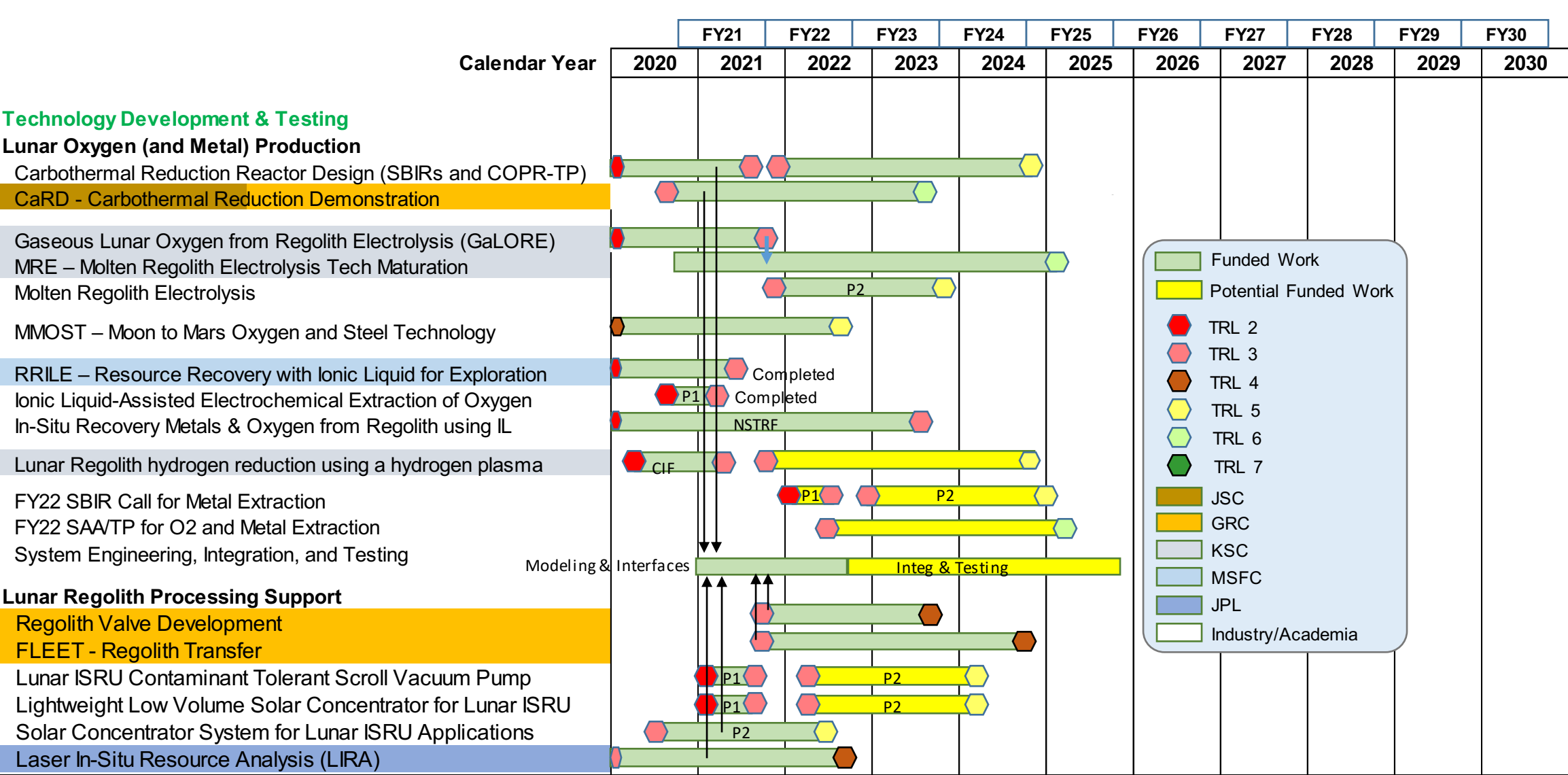
Regolith Excavation, Handling, & Manipulation Capability Gaps

- Long-life, regolith transfer (100s of mT) and low leakage regolith inlet/outlet valves for processing reactors (10s of thousands of cycles)
- Excavation and delivery of granular regolith (O₂/Metal) and icy regolith (Water Mining) – [Autonomous Excavation, Construction, & Outfitting \(AECO\)](#)
- Extensive Traversability (100s of km in sunlit and PSR locations and ingress/egress – [Autonomous & Robotic Systems Gap](#)

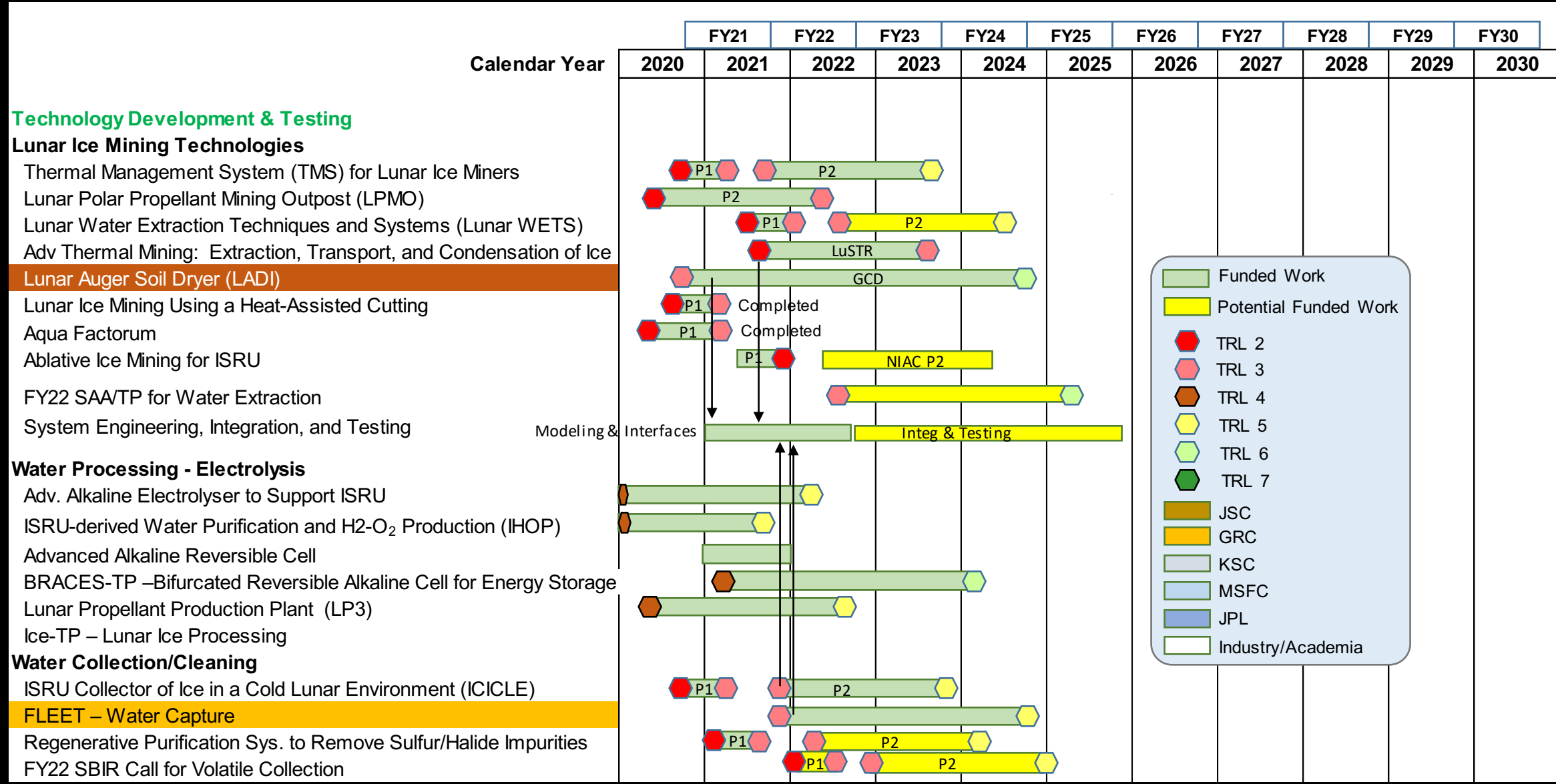
Cross-Cutting/System Level Resource Gaps

- Gravity-related research (short duration & ISS) to better understand impact on material handling, resource processing, and feedstock behavior
- Long-duration (100s of days) and Industrial-scale (10s of mT) operations under lunar vacuum and at <100 K temperatures
- Sensors and autonomous process monitoring and operations
- Industrial-scale water electrolysis, clean-up, and quality measurement for electrolysis or drinking (10s of mT/yr)

Current Oxygen-Metal from Regolith Development



Current Water Mining and Processing Development



ISRU Commodity Production Investment Status (1 of 2)



▪ Develop Critical Technologies for Lunar Oxygen Extraction

- ✓ Close coordination with Autonomous Excavation, Construction, and Outfitting (AECO) on excavation and delivery
- ✓ 6 different O₂ extraction technologies in development
- ✓ 9 development projects for 3 different water electrolysis approaches (with Life Support and Regenerative Power)
- Interface and internal technologies/functional areas require further investment

▪ Develop Critical Technologies for Lunar Resource Assessment and Water Extraction

- ✓ Significant number of SMD and STMD instrument technologies for resource assessment down to 1 m.; University/Public Challenges
- ✗ Need to consider technologies for deeper >3 m assessment for water/volatiles based on some water deposit theories
- ✓ Close coordination with AECO on excavation in Permanently Shadowed Regions (PSRs); Break the Ice Lunar Challenge
- 6 water mining development projects for 3 different approaches
- ✓ 9 development projects for 3 different water electrolysis approaches (with Life Support and Regenerative Power)
- Interface and internal technologies/functional areas require further investment
- ✗ No dedicated robotic polar water/volatile resource assessment surface missions beyond VIPER currently in planning
- ✗ No dedicated funded effort to develop resource maps for site selection

▪ Develop Critical Technologies for Manufacturing and Construction Feedstocks/Commodities

- Technologies for raw metal/alloy extraction in work as part of O₂ extraction; work required to further separate and refine metals
- Technologies for regolith size sorting, mineral beneficiation, and regolith manipulation in work
- Development and evaluation feedstocks to support manufacturing and construction techniques
- ✗ Limited plastic/binder production from in situ resources; synthetic biology technologies in work for bio-plastic and some commodity feedstocks

▪ Evaluate and Develop Integrated Systems for Extended Ground Testing; Tie to Other Discipline Plans

- ✓ NASA and APL performed/performing ISRU system evaluations
- Dedicated modeling, evaluation criteria, and Figures of Merit (FOMs) established
- Approach/approval for NASA and/or Industry-led System development and testing
- ✓ Facilities and simulants to support lunar environmental testing with regolith simulants
- Facilities and approach for extended mission analog operation and evaluation ground testing

Green = Significant Funded Activities
Yellow = Partially Covered; More Required
Red = Limited/No Funded Activities

ISRU Commodity Production Investment Status (2 of 2)



- **Develop/Fly Resource Assessment & ISRU Demonstrations Missions leading to Pilot Plant operations by 2030**
 - ☑ Orbital missions, PRIME-1, & VIPER funded and under development for launch
 - ☐ Lunar Trailblazer launch date and mission data later than desired. Actual spacecraft ready for launch in 2022
 - ☒ No clear plan for polar water/volatile resource assessment leading to Base Camp site selection – predicated on success of VIPER
 - ☐ At least one demonstration planned for each ISRU commodity path
- **Involve Industry/Academia with Goal of Commercial Space Operations at Scale**
 - ☑ 25, NIACs, SBIRs, BAAs, ACOs, & TPs led by industry underway for ISRU
 - ☑ 9 STTRs, NIACs, LuSTR, NSTRF, ESI/ECF led by Academia underway for ISRU
 - ☑ Lunar Surface Innovation Consortium – ISRU Focus Group underway and active; Supply/Demand Workshop
 - ☑ Center for the Utilization of Biological Engineering in Space (CUBES)
 - ☑ NASA prize competitions and university challenges: BIG Idea, Moon-Mars Ice Prospecting, Break the Ice Lunar, Lunabotics, CO₂ Conversion Challenge, Space Robotics Challenge
 - ☐ Selection/Competition strategy for ISRU demonstrations and Pilot Plant in work for industry involvement and commercialization

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ISRU Commodity Production Summary and Next Step Priorities



- **Complete Development of Water/Oxygen Mining Paths and Close Technology Gaps**
 - Continue oxygen extraction of Highland regolith
 - Continue water extraction/mining approaches in parallel until mission data allows for down-selection
 - Work with life support on oxygen and water cleanup technologies and requirements
- **Expand Development of Metal/Aluminum Extraction & other Feedstock for Manufacturing & Construction**
 - Continue and expand work on combined oxygen and metal extraction technologies;
 - Initiate work focused on metal extraction and processes leading to more pure/refined metals
 - Consider wider range of regolith options: Mare regolith, Pyroclastic Glasses, and KREEP
 - Continue and expand construction feedstock/commodity development with in-space manufacturing and construction
 - Evaluate synthetic biology technologies for bio-mining, bio-plastic, and some commodity feedstocks
- **Coordinate Polar Resource Assessment with SMD and ESD/SOMD for Artemis Base Camp site selection**
- **Initiate Internal and Industry-led System-level integration of ISRU and infrastructure capabilities**
 - Expand ISRU system engineering, modeling, integration, and testing to enable technology and system selections
 - Begin combining power, excavation, ISRU, storage & transfer, comm/nav, autonomy/avionics, maintenance/crew.
- **Initiate solicitations with Industry to progress ISRU technologies to Demonstration & Pilot-scale flights**
 - Pursue oxygen and metal extraction demonstrations; delay water mining demonstration until better knowledge is obtained
 - Provide feedstock technologies and capabilities to support construction demonstrations



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